

Pittsburgh 311 Streets and Mobility Corpus

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Context and purpose

Pittsburgh's 311 system handles about 104,000 requests per year. The City's own 311 User Guide reports that 5 to 10 percent of requests filed through the web form get placed in the wrong category by the resident who files them. Miscategorized requests end up at the wrong department, which delays service and adds work for staff who have to re-route them. The larger group project is an LLM tool that reads a resident's complaint and suggests the correct category and department before the request enters the queue. This assignment is the Streets and Mobility slice of the data foundation for that tool.

What the corpus contains

Record type	Count	Source
Real filed 311 requests, 2015 to 2017 archive	70	WPRDC (real, tabular)
Resident complaints, derived from real requests	115	Authored (traceable to a real request_id)
Routing knowledge cards	5	Authored, grounded in real routing

Tabular share sits at 36.8 percent, well above the 20 percent minimum. Every authored complaint carries the request_id of the real filed request it was built from, which means each label in the answer key is what the City actually did with that specific real request. Two registers were used for complaints: 69 in a web-form style (short, informal, sometimes with typos) and 46 in an operator-note style (neutral, third-person).

Why the complaints had to be authored

The public 311 datasets publish structured fields only. Request type, department, neighborhood, timestamps, submission channel. There is no column with the resident's original wording. Whatever the caller said on the phone or typed into the web form was collapsed to a dropdown selection and the original text was never released. To test whether a language model can classify complaint text, complaint text had to be written. Each authored complaint was tied to one real filed request so its correct answer is grounded in a real routing decision the City made.

Human versus Phi results

The evaluation set is 25 complaints, 5 per category, each labeled with the category and department that the City actually assigned to the parent request. Phi received only the complaint text and the field menu.

Baseline (v1) results as scored:

Field	Measured accuracy
Category	0%
Department	0%
Entities	0%
Schema violation rate	100%
Mean latency per record	7.34 s

Reading the raw outputs told a different story. Phi was wrapping every answer in a JSON Schema structure instead of returning flat JSON. The grading script could not parse this, so it counted every field as wrong. When the values were pulled out of the wrapper by hand:

Field	Real v1 accuracy
Category	68% (17 of 25)
Entities	92% (23 of 25)

Department	0% (0 of 25, this one was really zero)
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Phi understood the task at the category and entity level. Its category answers were correct two thirds of the time. Its entity extraction was near-perfect. But it defaulted to the string "Public Works" for the department field in 17 out of 25 cases, because it had no way to know that Pittsburgh uses specific unit names like "Allegheny City Electric" or "DOMI - Traffic."

Two failures were happening at once. A format compliance failure that hid Phi's real accuracy, and a domain knowledge failure on the department field.

Recovery strategy

The revised prompt (v2) targeted both problems. First, it included an example of the exact flat JSON format expected, with an explicit instruction not to wrap the output in schema keywords. Second, it listed all 16 canonical department strings and the routing rules that map request types to departments. Streetlight repair goes to Allegheny City Electric. Illegal parking goes to Police Zones 1 through 6. Abandoned vehicles go to Police AVU. And so on.

Rerunning the same 25 records with v2:

Field	v1 measured	v2 measured
Category	0%	84%
Department	0%	40%
Entities	0%	96%
All fields correct	0 of 25	6 of 25
Schema violation	100%	0%

The recovery prompt did two jobs. The format example fixed a small-model compliance problem that had been hiding real accuracy. The routing rules gave Phi domain knowledge it did not otherwise have. Category jumped from 0 to 84 percent (mostly format recovery plus some rule application). Entities jumped from 0 to 96 percent (pure format recovery). Department climbed from truly zero to 10 correct out of 25, which is the field where domain knowledge actually mattered.

The 15 records Phi still gets wrong on department reveal a real pattern. Phi over-applies the routing rules once it has them. It sends sign and signal complaints to "DOMI - Traffic" even when the City actually routes them to "Permits, Licenses and Inspections." It defaults streetlight complaints to "Allegheny City Electric" even for new-installation requests that should go to "DOMI - Permits." So the routing rules helped, and hurt in a specific way when the rule did not apply.

Cluster reading

TF-IDF representations of all 190 records were grouped with K-means at k=6. Silhouette score was 0.046.

Cluster	n	Top terms	Reading
0	39	ave, streetlight, sign, resident, read	Web-form complaints about lights and signs
1	38	request, domi, routed, archive, record	Real archive rows routed to DOMI units
2	29	status, assigned, filed, closed, permits	Real archive rows routed to PLI, closed status
3	24	theres, issue, help, sidewalk, steps	Sidewalk and steps complaints, web-form voice
4	23	light, street, repair, allegheny, electric	Streetlight repair cluster (Allegheny City Electric)

5	37	reports, caller, constituent, vehicle, meter	Operator-note complaints, mostly parking
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The clusters split on two things at once: what the complaint is about (streetlights, sidewalks, parking) and how the record is written (real archive row, web-form voice, operator-note voice). The low silhouette reflects that short civic text uses heavy shared vocabulary, so cluster boundaries are soft. This is a real property of the domain, not a corpus flaw.

The cluster reading connects to the Phi results in a useful way. Clusters 4 and 0, which have distinctive vocabulary tied to specific routing rules, are where Phi's v2 recovery gets most of its correct answers. The Permits, Licenses and Inspections cases, which are scattered across several clusters and never signaled by distinctive vocabulary, are where recovery still fails. So a cluster whose top terms match a routing rule tends to be one Phi handles well after v2.

Bonus: Phi versus Claude Sonnet 4.6

Same 25 records, same v1 prompt (no routing rules, no format example), run through Claude Sonnet via OpenRouter.

Field	Phi v1 measured	Phi v2 recovery	Claude v1
Category	0% (real 68%)	84%	92%
Department	0%	40%	4%
Entities	0% (real 92%)	96%	100%
Schema violation	100%	0%	0%
Mean latency	7.34 s (local)		1.56 s (hosted)

Two findings. Model size fixed the format compliance problem for free: Claude produced valid JSON on every record without being told how, while Phi needed an explicit example. On category and entities, Claude outperformed Phi as expected.

The interesting result is the department. Claude scored 4 percent (1 of 25) on the department field, worse than Phi with the recovery prompt at 40 percent. Claude defaults to generic names like "Public Works" and "Department of Public Works" because it has no way to know Pittsburgh uses "DOMI - Traffic" or "Allegheny City Electric" unless we tell it. A four-billion-parameter local model with the routing rules beat a frontier hosted model without them, on the hardest field. Domain knowledge in the prompt matters more than model size when the answer set is specific to a jurisdiction.

Limitations

Complaints are authored, one per real filed request. The routing decisions are real but the words are ours. Source concentration is 97 percent one domain (WPRDC), inherent to the design. Median document length is 92 characters, because civic text is terse, which limits available context. The 2015 to 2017 window straddles the DPW to DOMI renaming, so the same work appears under two department name variants. The evaluation used the 25 human labels the assignment requires; more labels would tighten the accuracy estimates.

Sources

City of Pittsburgh via WPRDC, 311 Data Archive 2015 to 2017 request log (Creative Commons Attribution). City of Pittsburgh 311 program pages, used for routing knowledge cards. Full URLs, retrieval date, and per-source record counts are in sources.csv.

The main takeaway from this evaluation is that for jurisdiction-specific routing, giving a small model the right domain rules closes more of the accuracy gap than switching to a model twenty times its size.